

GIS Mainpage / Archived Local Storm Reports Shapefiles

This page allows you to generate a GIS shapefile of NWS issued Local Storm Reports for a time period of your choice. Please be sure to read the archive status and dbf schema below. The IEM's archive is certainly not complete or official.

If you are looking for a real-time source of lsrs, you can find with the following files generated every 5 minutes: ESRI Shapefile, Comma Delimited, GeoJSON.

Looking to automate the downloading of LSRs?

You can use the following URI form to get the most recent number of seconds worth of LSRs. Note that these times are based on the valid time of the LSR and not the issuance of the text product itself that contained the LSR.

https://mesonet.agron.iastate.edu/cgi-bin/request/gis/lsr.py?wfo=ALL&recent=86400&fmt=csv
https://mesonet.agron.iastate.edu/cgi-bin/request/gis/lsr.py?wfo=DMX,DVN&recent=86400&fmt=csv
https://mesonet.agron.iastate.edu/cgi-bin/request/gis/lsr.py?state=IA&recent=86400&fmt=csv

Omit the fmt=csv portion to return the zipped shapefile.

Example to fetch a month's worth of data

https://mesonet.agron.iastate.edu/cgi-bin/request/gis/lsr.py?wfo=JAN&sts=2020-08-01T00:00Z&ets=2020-09-01T00:00Z&fmt=csv

 Backend documentation

 exists for those wishing to script against this service.

Limit Weather Forecast Office (WFO)

All Available
[ABR] Aberdeen
[ALY] Albany
[ABQ] Albuquerque
[AMA] Amarillo

Limit by State

All Available
[AL] Alabama
[AK] Alaska
[AR] Arkansas
[AS] American Samoa

Limit by Report Type

[SNOW SQUALL] Snow Squall
[STORM SURGE] Storm Surge
[TORNADO] Tornado
[TROPICAL CYCLONE] Tropical Cyclone

Select time interval

(Times are in UTC. Local Storm Reports with an event time **valid** between these times selected below will be included in the output GIS file...)

	Year	Month	Day	Hour	Minute
Start:	2020	Jan	1	0 UTC	0
End:	2024	Dec	31	23 UTC	59

Download Format:

ESRI Shapefile

Giveme File Now Please!

Shapefile DBF schema:

VALID	Timestamp of LSR in GMT/UTC time
MAG	Magnitude value of the LSR
WFO	Weather Forecast Office originating the LSR
TYPECODE	1 character identifier of the report type, IEM specific
TYPETEXT	Textual value of LSR type used in report
CITY	Location used for the LSR
COUNTY	County/Parish of the LSR city
STATE	Two character state abbreviation
SOURCE	Who reported the LSR
REMARK	Text summary with the LSR
LAT	Latitude
LON	Longitude
UGC	IEM computed NWS UGC code (6 character), sometimes null
UGCNAME	IEM computed NWS UGC name (128 character), sometimes null
QUALIFY	1 char describing if the MAG is Estimated, Measured, or Unknown

Archive notes:

- Before 1 Dec 2006, there was no differentiation between Snow and Sleet reports.
- 18 July 2022:** Added output fields that attempt to cross reference NWS UGC County/Zones database to the referenced county/state found in the LSR text. This all is sub-optimal, but works in 99% of cases.

Publications Citing IEM Data (View All)

These are the most recent 10 publications that have cited the usage of data from this page. This list is not exhaustive, so please let us know if you have a publication that should be added.

- Sabol, A., C. Nowotarski. 2025, Evolution of Southern Plains Supercells and Tornadoes through the Early Evening Transition. J. Operational Meteor., 13 (1), 1-17 <https://doi.org/10.15191/nwajom.2025.1301>
- Bower, E., M. Erickson, et al. 2024, Objective Verification of the Weather Prediction Center's Mesoscale Precipitation Discussions. Weather and Forecasting. Volume 39, Issue 6. <https://doi.org/10.1175/WAF-D-23-0199.1>
- Hill, A. and R. Schumacher. 2021, Forecasting Excessive Rainfall with Random Forests and a Deterministic Convection-Allowing Model. Weather and Forecasting. Volume 36, Issue 5. <https://doi.org/10.1175/WAF-D-21-0026.1>
- Hill, A., C. Weiss, et al. 2021, Influence of a Portable Near-Surface Observing Network on Experimental Ensemble Forecasts of Deep Convection Hazards during VORTEX-SE. Weather and Forecasting. Volume 36, Issue 4. <https://doi.org/10.1175/WAF-D-20-0237.1>

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